

Assignment-1

Name - Savadiya Paul

Roll no. - 51

Univ Roll - 12200222049

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Tokenization is a fundamental preprocessing step in Natural Language Processing (NLP) that involves breaking a given text into smaller units called tokens. These tokens can be sentences, words, sub-words, or characters depending on the requirement of the task. Since computers cannot directly understand raw human language, tokenization converts unstructured text into a structured form that machine learning and NLP algorithms can process effectively.

The primary role of tokenization is to establish the basic units for further linguistic analysis such as stemming, lemmatization, part-of-speech tagging, and ~~syntactic~~ syntactic parsing. For eg, the sentence "Natural Language Processing is interesting" can be tokenized into individual words

like Natural, Language, Processing, is and interesting. These tokens act as inputs for features extraction techniques such as Bag of Words (Bow) and TF-IDF.

Tokenization also helps in reducing ambiguity and noise in text. Proper handling of punctuation, special characters, numbers, and contractions (e.g., don't → do + not) improve the accuracy of downstream NLP tasks. Advanced tokenization are particularly useful in handling unknown words, spelling variations, and morphologically rich languages.

Overall, tokenization plays a crucial role in ensuring that textual data is accurately represented, computationally efficient, and suitable for tasks like text classification, sentiment analysis, machine translation, and information retrieval.

2.

	<u>Syntax</u>	<u>Semantics</u>	<u>Pragmatics</u>
Meaning	Deals with the structure of sentences.	Deals with the literal meaning of sentences.	Deals with meaning based on context.
Focus	Word order & grammatical rules.	Meaning of words and sentences.	Speaker intention and situational context.
Concern	How a sentence is formed.	What a sentence means.	What a sentence intends to convey.
Dependency on Context	Does not depend on context.	Most independent of context.	Highly dependent on context.
Ambiguity Handling	Cannot resolve ambiguity	Partially resolves ambiguity	Resolves ambiguity using context.
Level of Analysis	Structural level	Meaning level.	Contextual level.
Example	"The boy eats an apple" Grammar checking	Understanding what "apple" means. word sense interpretation	Interpreting whether it is a request, command, or statement.
Role in NLP	Helps in parsing sentences.	Helps in dialogue systems meaning representation	Helps in dialogue systems.

3.

Before applying K-NN, textual documents must be converted into numerical form. Common techniques include Bag of Words (Bow) and TF-IDF (Term Frequency - Inverse Document Frequency). Each document is represented as a vector in a high-dimensional feature space where each dimension corresponds to a unique term.

Working Principle:-

- Training phase— KNN does not build an explicit model. All labeled documents are stored in the training dataset.
- Similarity Measurement → When a new document is given, it is converted into a feature vector. The similarity between this document and all training documents is calculated using measures such as cosine similarity or Euclidean distance. Cosine similarity is preferred for text as it handles high-dimensional sparse data efficiently.
- Selection of k nearest neighbors → The top k documents with the highest similarity scores are selected as nearest neighbors.
- Classification → The class label is assigned to the new document based on majority voting among the k nearest neighbors.

Eg:- Consider documents categorized as Sports and Politics.

If a new document related to "cricket match tournament" is most similar to documents labeled as Sports, then k -NN assigns the class Sports to the new document.